

The Influence of Social Vulnerability and Culture on Physical Inactivity in the United States — Identifying Hot Spots in Need of Attention

Ross Arena, PhD,^{a,b} Nicolaas P. Pronk, PhD,^{b,c,d} Colin Woodard, MA, FRGS^e

^aDepartment of Physical Therapy, College of Applied Science, University of Illinois, Chicago; ^bHealthy Living for Pandemic Event Protection (HL — PIVOT) Network, Chicago, Ill; ^cHealthPartners Institute, Minneapolis, Minn; ^dDepartment of Health Policy and Management, University of Minnesota, Minneapolis; ^eNationhood Lab, Pell Center for International Relations and Public Policy, Salve Regina University, Newport, RI.

ABSTRACT

BACKGROUND: In the United States, ongoing efforts to increase individual- and population-level physical activity have made little to no progress over the past decade in reducing the percentage of individuals who report no leisure-time physical activity. The purpose of the current study is to further assess the relationship between social vulnerability and the prevalence of physical inactivity at the county level.

METHODS: We merged county-level data on physical inactivity prevalence, the Social Vulnerability Index (SVI), and the American Nations regional cultures schematic.

RESULTS: Physical inactivity significantly correlated ($P < .001$) with the overall SVI score and all sub-theme scores to varying levels of strength. Clear and statistically significant heterogeneity in the SVI scores was apparent across distinct regions using the American Nations model, consistent with previously demonstrated patterns of physical inactivity prevalence.

CONCLUSIONS: On a national level, physical inactivity prevalence is unacceptably high and has not appreciably improved over the past decade. Within the United States, high levels of social vulnerability and physical inactivity are concentrated within specific geographic regions that need tailored solutions to resolve health disparities.

© 2023 Elsevier Inc. All rights reserved. • *The American Journal of Medicine* (2023) 000:1–9

KEYWORDS: Chronic disease; Culture; Health equity; Moving more sitting less; Prevention; Sedentary lifestyle

INTRODUCTION

Being physically active promotes healthy longevity, prolonging both healthspan and lifespan.^{1–4} In the United States, ongoing efforts to increase individual- and population-level physical activity have made little to no progress over the past decade.⁵ The coronavirus disease 2019

(COVID-19) pandemic may have worsened the situation, as social distancing and stay-at-home orders to contain the virus caused people to move even less.^{6–9} We are now watching to see if the lower COVID-19-era physical activity levels persist. In either case, the physical inactivity crisis itself represents an ongoing pandemic.^{10,11}

The Healthy People 2030 initiative has set a goal of reducing the “proportion of adults who do no physical activity in their free time” to 21.8%—a 4.3% reduction from 2020 levels nationally.¹² Using a more detailed, county-level view, the United States is clearly heterogeneous in terms of the distribution of physical inactivity prevalence. We have recently demonstrated the heterogeneity in this pattern using the *Americans Nations*¹³ model,¹⁴ identifying regions that are physical inactivity “hot spots.” Given the heterogeneous county-level distribution, there is a need

Declaration of Competing Interest: The authors have no disclosures or conflicts regarding this submission.

Funding: None.

Authorship: All authors had access to the data. RA prepared the initial draft of the manuscript. RA and CW prepared data analysis. NPP provided critical revisions and new content to the manuscript draft.

Requests for reprints should be addressed to Ross Arena, PhD, Department of Physical Therapy, College of Applied Health Sciences, University of Illinois Chicago, 1919 W. Taylor Street (MC 898), Chicago, IL 60612.

E-mail address: raarena@uic.edu

to identify factors that are meaningfully associated and perhaps predictive of physical inactivity on a local level.

There are several factors that reflect an increased likelihood of social vulnerability that may reduce physical activity, including disability, unfavorable socioeconomic conditions, and limited access to health care.¹⁵⁻¹⁸ The purpose of the current study is to further assess the relationship between social vulnerability and the prevalence of physical inactivity at the county level. A second objective is to provide a cultural geographic overlay of this relationship using the American Nations model.

METHODS

Physical Inactivity

We obtained 2021 county-level, age-adjusted, “no leisure time physical activity” prevalence from the 2023 Centers for Disease Control and Prevention (CDC) PLACES database.¹⁹ Data on “no leisure time physical activity” were obtained through the Behavioral Risk Factor Surveillance System survey.²⁰ Physical inactivity was also coded (0 and 1) at the county level using a $\leq 21.8\%$ physical inactivity threshold—the national goal according to Healthy People 2030.¹²

Social Vulnerability

The 2020 CDC/Agency for Toxic Substances and Disease Registry (ATSDR) Social Vulnerability Index (SVI) database provides overall and subtheme SVI scores.²¹ The SVI attempts to capture the potential negative effects of external health stressors using 16 US census variables (Figure 1).²¹ There are 4 SVI subthemes, calculated as the sum of percentages (0-1.0) for each constituent variable. Overall, SVI is the sum of the 4 subthemes, and higher scores indicate greater social vulnerability. Data used to calculate the 2020 CDC/ATSDR SVI were derived from 5-year (2016-2020) American Community Survey data.²²

Regional Cultures

The American Nations regional model was obtained from the Nationhood Lab. This model has been described elsewhere,¹³ but briefly, it discerns and maps North American regional cultures based on “first settler effects”²³ and competing colonization patterns, which often do not respect state or even international boundaries. The CDC PLACES, CDC/ATSDR SVI, and Nationhood Lab databases were linked through ZIP Code identifiers.

Subject Protection

HealthPartners Institute Research Subjects Protection Program determined that this study is exempt from institutional review board review and ongoing oversight under 45 CFR Part 46, as it involves the analysis of existing, publicly available data sets.

CLINICAL SIGNIFICANCE

- In the United States, ongoing efforts to increase physical activity levels have made little to no progress over the past decade.
- Clear and statistically significant heterogeneity in social vulnerability is apparent across distinct regions using the American Nations model.
- Within the United States, high levels of social vulnerability and physical inactivity are concentrated within specific geographic regions that need tailored solutions to resolve health disparities.

Statistical Analysis

Pearson’s correlation assessed the relationships between SVI overall and subtheme scores and physical inactivity on a county level. Stepwise linear regression analysis (entry = 0.05, removal = 0.10) was used to assess the ability of SVI subtheme scores and American Nations dichotomous grouping to predict US County physical inactivity as a continuous variable. Receiver operating characteristic (ROC) curve analysis was used to further assess the ability of the SVI to differentiate counties with a $\leq 21.8\%$ physical inactivity prevalence and identify optimal SVI threshold values based on sensitivity and specificity. County-level overall SVI score was also visually

depicted according to the American Nations model to identify areas with particularly high SVI scores that warranted further region-specific analyses. Independent *t* tests were used to compare: 1) SVI scores between counties using a $\leq 21.8\%$ physical inactivity threshold; and 2) SVI scores and physical inactivity prevalence by American Nation dichotomous grouping. Chi-square analysis was used to assess differences in distribution of $\leq 21.8\%$ physical inactivity prevalence according to American Nation dichotomous grouping. All statistical tests with a *P* value $< .05$ was considered significant. SPSS (Version 28, IBM, Armonk, NY) was used for all statistical analyses.

RESULTS

Data on SVI and physical inactivity prevalence was available for 3148 and 3072 counties, respectively. Table 1 lists correlations between physical inactivity and SVI subtheme scores as well as overall score. Physical inactivity significantly correlated with all subtheme scores and the overall SVI score to varying levels of strength. Stepwise linear regression analysis revealed: 1) Socioeconomic Status Subtheme Score; 2) Racial and Ethnic Minority Status Subtheme Score; and 3) Household Characteristics Subtheme Score were all retained (all *P* $< .001$) in the linear regression predicting physical inactivity as a continuous variable (*R* = 0.69, *R*² = 0.47). Housing Type and Transportation Subtheme Score was removed from the final multivariate regression (*P* = .15).

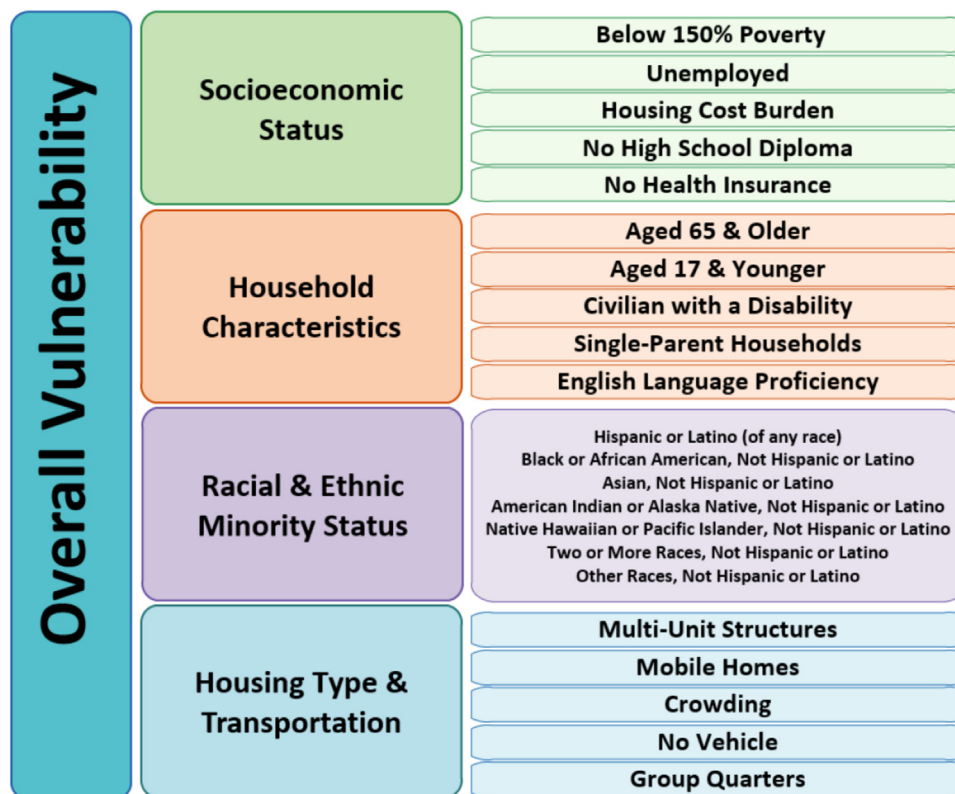


Figure 1 Variables included in the Centers for Disease Control and Prevention (CDC)/Agency for Toxic Substances and Disease Registry (ATSDR) Social Vulnerability Index.

Source: CDC SVI Documentation 2020.²¹ https://www.atsdr.cdc.gov/placeandhealth/svi/documentation/SVI_documentation_2020.html. Measures derived from derived from 5-year (2016-2020) American Community Survey data.

The regression equation for predicting physical inactivity prevalence was:

Percent Physical Inactivity

$$= 15.271 + 3.271(\text{Socioeconomic Status}) \\ - 4.279(\text{Racial and Ethnic Minority Status}) \\ + 2.196(\text{Household Characteristics}).$$

Physical Inactivity/SubthemeScores : 00.00

Only 512 (17%) of the 3072 US counties had a physical inactivity prevalence <21.8%. Comparatively, physical inactivity prevalence ($28.1 \pm 4.3\%$ vs $19.1 \pm 2.1\%$ years), Socioeconomic Status Subtheme Score (2.6 ± 0.98 vs 1.7 ± 0.54 vs 2.0 ± 0.52), Household Characteristics Subtheme Score (2.5 ± 0.68 vs 2.2 ± 0.70) and overall SVI score (8.2 ± 2.0 vs 6.4 ± 1.6) were all higher in the counties with a physical inactivity prevalence $\geq 21.8\%$ (all $P < .001$). Conversely, Racial and Ethnic Minority Status Subtheme Score (0.50 ± 0.30 vs 0.48 ± 0.23) was not significantly different between counties using the $</\geq 21.8\%$ physical inactivity prevalence threshold ($P = .29$).

Results for ROC analysis are listed in Table 2. Area under the curve was significant for 3 of the 4 subtheme

scores and the overall SVI score; threshold values with optimal sensitivity and specificity were identified.

Figure 2 illustrates the geographic distribution of the overall 2020 SVI according to the American Nations regional cultures model, both as an average SVI in each region (Figure 2A) and at the county level (Figure 2B)—clear heterogeneity in SVI scores exists across regions. In a previous publication assessing physical inactivity

Table 1 Pearson Product Moment Correlation Results

	Prevalence of No Leisure-Time Physical Activity
Socioeconomic Status Subtheme Score	0.64*
Household Characteristics Subtheme Score	0.50*
Racial and Ethnic Minority Status Subtheme Score	0.20*
Housing Type and Transportation Subtheme Score	0.33*
Overall Social Vulnerability Index	0.59*
* $P < .001$.	

Table 2 Receiver Operating Characteristic Curve Analysis

Area Under the Curve					
Test Result Variable(s)	Area	P Value	Optimal Threshold Value	Asymptotic 95% Confidence Interval	
				Lower Bound	Upper Bound
Socioeconomic Status Subtheme Score	0.77	< .001	$</\geq 2.10$ (Sen: 70%/Spec: 70%)	0.75	0.79
Household Characteristics Subtheme Score	0.75	< .001	$</\geq 2.30$ (Sen: 67%/Spec: 70%)	0.73	0.77
Racial and Ethnic Minority Status Subtheme Score	0.51	.44	$</\geq 0.62$ (Sen: 39%/Spec: 74%)	0.49	0.53
Housing Type and Transportation Subtheme Score	0.62	< .001	$</\geq 2.18$ (70%/Spec: 47%)	0.59	0.64
Overall SVI Score	0.75	< .001	$</\geq 7.20$ (68%/Spec: 69%)	0.73	0.77

Sen = sensitivity; Spec = specificity; SVI = Social Vulnerability Index.

prevalence according to the American Nations model, Greater Appalachia, the Deep South, El Norte, the Spanish Caribbean, and First Nation demonstrated a significantly higher physical inactivity prevalence compared with other regions (Figure 3).¹⁴ Table 3 lists correlations between physical inactivity and SVI subtheme scores as well as overall score in the combined Greater Appalachia, Deep South, El Norte, Spanish Caribbean, and First Nation regions. Strength of correlations in the subgroup analysis (Table 3) was comparable with results from the overall group (Table 1). Table 4 lists correlations between physical inactivity and SVI subtheme scores as well as overall score in the combined Yankeedom, Midlands, New Netherland, the Federal Entity, Tidewater, Far West, Left Coast, and Greater Polynesia regions. The correlations, while significant, were overall weaker than the overall group and other American Nations subgrouping. Table 5 compares subtheme and overall SVI scores as well as physical inactivity prevalence according to dichotomous American Nations grouping: Group 1 = Yankeedom, Midlands, New Netherland, the Federal Entity, Tidewater, Far West, Left Coast, and Greater Polynesia; and Group 2 = Greater Appalachia, Deep South, El Norte, Spanish Caribbean, and First Nation. All SVI subtheme scores, the overall SVI score, and physical inactivity prevalence were significantly higher in Group 2. The percentage of counties with a physical inactivity prevalence $<21.8\%$ in Group 1 and Group 2 was 28.7% and 4.1%, respectively, which was significantly different ($P < .001$). The American Nations dichotomous grouping (ie, Group 1 vs Group 2) was then added to SVI subtheme scores to predict physical inactivity prevalence in a subsequent linear regression analysis. The second stepwise linear regression analysis revealed: 1) Socioeconomic Status Subtheme Score; 2) American Nation dichotomous grouping; 3) Racial and Ethnic Minority Status Subtheme Score; 4) Household Characteristics Subtheme Score; and 5) Housing Type and Transportation Subtheme Score were all retained (all $P < .01$) in the linear regression, predicting physical inactivity as a continuous variable ($R = 0.73$, $R^2 = 0.54$). The

regression equation for predicting physical inactivity prevalence was:

Percent Physical Inactivity

$$= 15.431 + 2.427(\text{Socioeconomic Status}) \\ + 3.061(\text{American Nations Dichotomous Grouping}) \\ - 3.880(\text{Racial and Ethnic Minority Status}) \\ + 1.962(\text{Household Characteristics}) \\ + 0.327(\text{Housing Type and Transportation}).$$

Physical Inactivity and Subtheme Scores in American Nations : 00.00

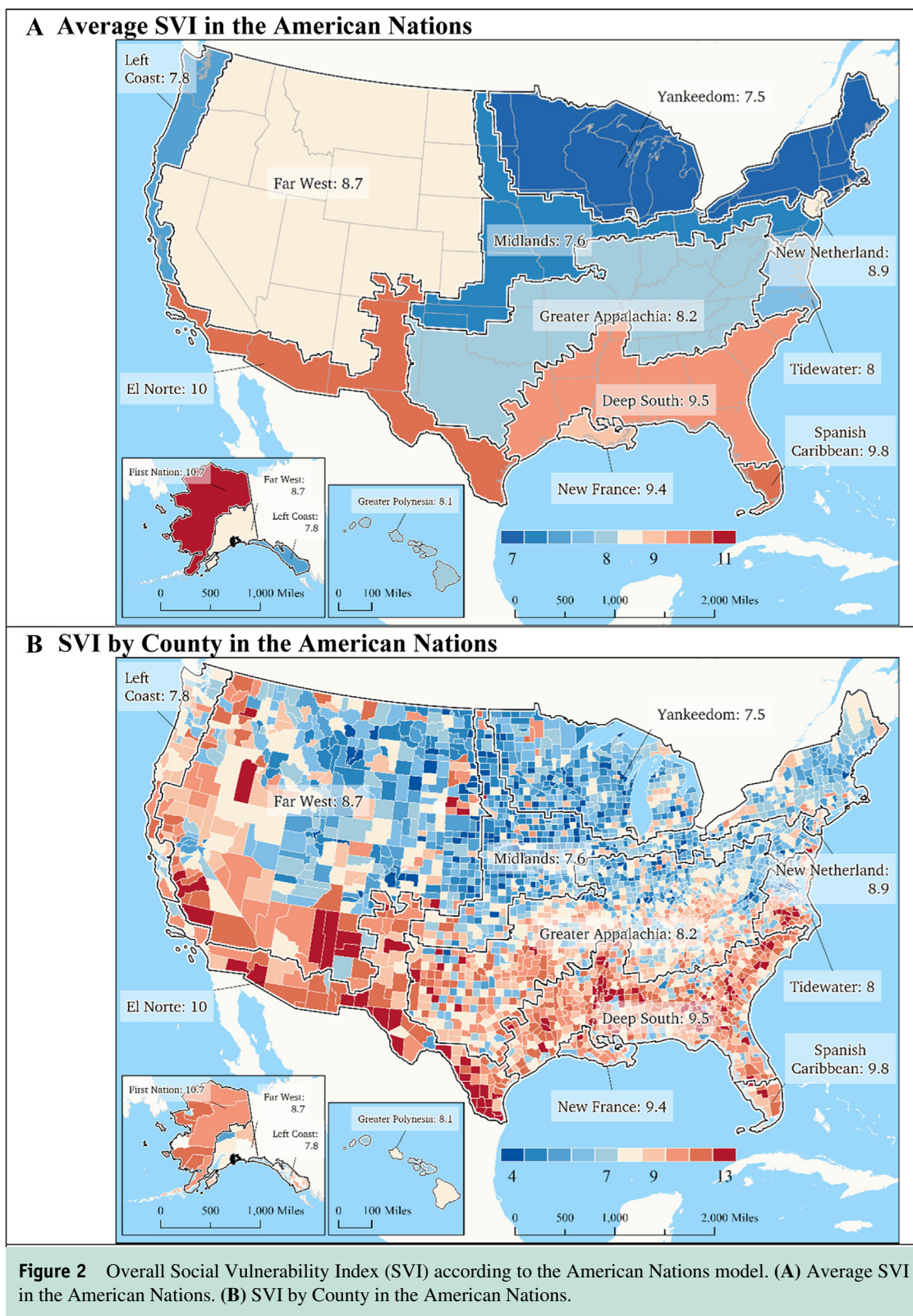
Group 1 = 0

Group 2 = 1

DISCUSSION

The United States has been in the throes of a physical inactivity pandemic¹¹ for some time, which may have worsened because of the COVID-19 pandemic,^{6,24} and it is unclear if physical activity will fully rebound.⁹ A prolonged post-pandemic re-calibration to an even lower level of physical activity will worsen the chronic disease crisis we face.^{25,26} Moreover, despite best intentions, we have made little progress in improving physical activity in the United States over the last decade.⁵ One of the reasons for the apparent lack of progress in addressing the physical inactivity health crisis is the approach to promoting physical activity from the individual and population level. Physical activity messaging and counseling often takes a “one-size-fits-all” approach. Moreover, the approach, when delivered, is often crafted in a dichotomous “all or none” fashion, focusing on ideal physical activity goals based on current guidelines (eg, 150 minutes or more of moderate physical activity on most or all days of the week).^{27,28} Historically, stakeholders committed to improving physical activity, from public health to individual health professionals, have not considered the true drivers of physical inactivity.

The results from the current analysis indicate: 1) Social vulnerability significantly influences physical inactivity



prevalence in the United States, especially when related to socioeconomic status and household characteristics; 2) The distribution of social vulnerability in the United States is heterogeneous—with the southern regions and Alaska demonstrating the highest SVI scores; 3) The distribution of the SVI and physical inactivity

prevalence across the United States overlap considerably, suggesting interactions between these adverse phenomena; and 4) As we previously demonstrated with physical inactivity prevalence,¹⁴ the distribution of SVI scores according to the American Nations model suggests a regional cultural influence.

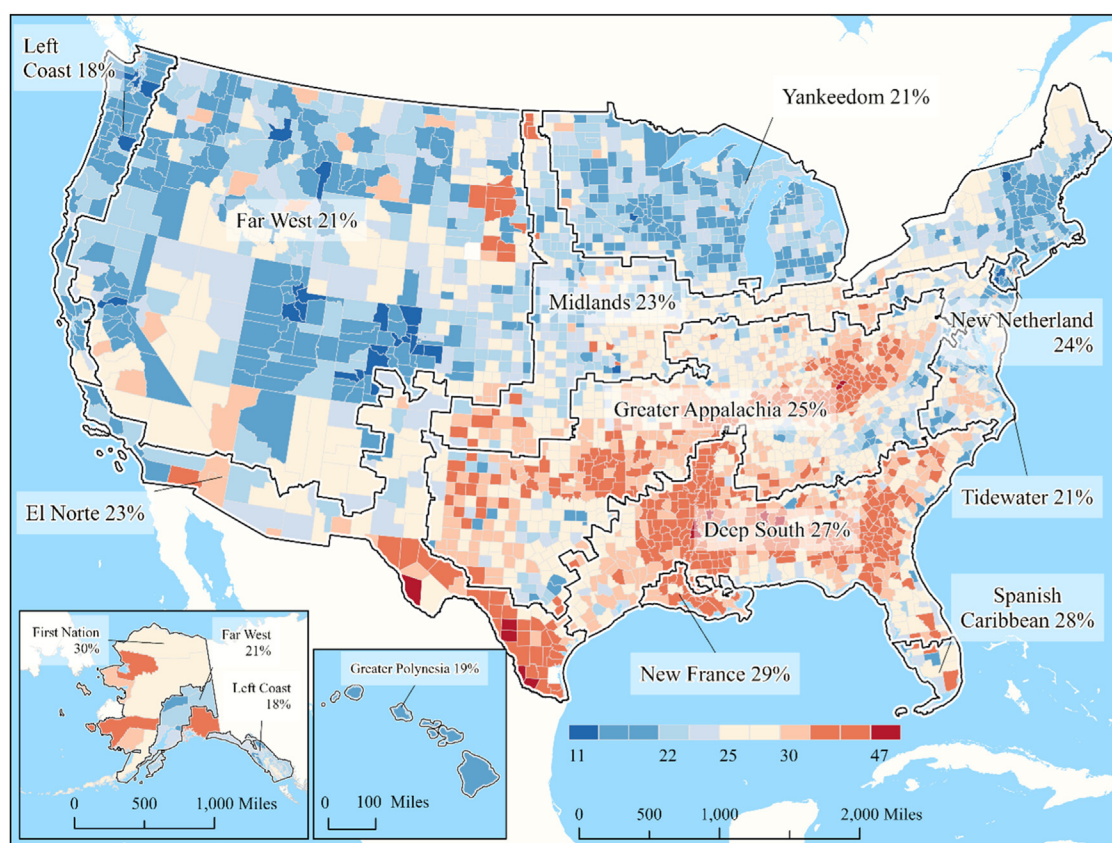


Figure 3 Physical inactivity prevalence by county according to the American Nations model.

Reprinted with Permission:¹⁴ Arena R, Laddu D, Pronk NP, Woodard C. The geographic distribution of unhealthy living characteristics according to the American Nations model: cultural factors warranting attention. *Prog Cardiovasc Dis.* 2023;79:100-106.

There have been previous investigations into the relationship between the SVI and myriad health indicators, including physical inactivity prevalence. This available previous evidence demonstrates a significant relationship between SVI subtheme scores and overall score with numerous health indicators, supporting the premise that increased social vulnerability influences both unhealthy lifestyle behaviors and health characteristics.^{15-18,29} The

findings of the current investigation corroborate previous findings and further support the importance of social vulnerability to identify regions/communities that are at a higher risk of unhealthy lifestyle behaviors, chronic disease, and poor health outcomes. Our findings are novel in several respects and thus, advance this area of inquiry as follows:

Table 3 Pearson Product Moment Correlation Results in the Combined Greater Appalachia, Deep South, El Norte, Spanish Caribbean, and First Nation Regions

	Prevalence of No Leisure-Time Physical Activity
Socioeconomic Status Subtheme Score	0.60*
Household Characteristics Subtheme Score	0.37*
Racial and Ethnic Minority Status Subtheme Score	0.15*
Housing Type and Transportation Subtheme Score	0.38*
Overall Social Vulnerability Index	0.55*

* $P < .001$.

Table 4 Pearson Product Moment Correlation Results in the Combined Yankeedom, Midlands, New Netherlands, Federal Entity, Tidewater, Far West, Left Coast, and Greater Polynesia Regions

	Prevalence of No Leisure-Time Physical Activity
Socioeconomic Status Subtheme Score	0.42**
Household Characteristics Subtheme Score	0.43**
Racial and Ethnic Minority Status Subtheme Score	0.07*
Housing Type and Transportation Subtheme Score	0.15**
Overall Social Vulnerability Index	0.39**

** $P < .001$.

* $P < .01$.

Table 5 Comparison of Overall and Subtheme SVI Scores and Prevalence of No Leisure-Time Physical Activity According to American Nations Grouping

	Group 1 1576 Counties	Group 2 1572 Counties
Socioeconomic Status Subtheme Score	1.99 ± 0.90	2.97 ± 0.86*
Household Characteristics Subtheme Score	2.27 ± 0.54	2.65 ± 0.53*
Racial and Ethnic Minority Status Subtheme Score	0.44 ± 0.27	0.56 ± 0.30*
Housing Type and Transportation Subtheme Score	2.31 ± 0.70	2.61 ± 0.64*
Overall SVI	7.02 ± 1.88	8.79 ± 1.81*
Prevalence of no leisure-time physical activity	23.79 ± 3.99%	29.60 ± 4.69%*

SVI = Social Vulnerability Index.

Group 1: Yankeeedom, Midlands, New Netherlands, Federal Entity, Tidewater, Far West, Left Coast, and Greater Polynesia; 1567 counties available for all comparisons.

Group 2: Greater Appalachia, Deep South, El Norte, Spanish Caribbean, and First Nation; 1572 counties available for SVI comparisons and 1505 counties available for physical activity comparison.

* $P < .001$.

1) The relationship between physical inactivity prevalence and both subtheme and overall SVI scores was assessed to determine specific associations and characterize the influence of specific components of social vulnerability in their ability to predict physical inactivity prevalence; 2) The threshold values for both subtheme and overall SVI scores that best predicted counties that were above or below the Healthy People 2030 goal for physical inactivity prevalence;¹² 3) The relationship between physical inactivity and social vulnerability was assessed in the context of unique regional cultures in the United States according to the American Nations model.¹³

Of the SVI components, those associated with socioeconomic status appeared to be most strongly related to physical inactivity prevalence on a county level as a continuous measure. Both socioeconomic status and household characteristics were comparably effective in classifying counties currently above and below the Healthy People 2030 physical inactivity prevalence goal.¹² Using these threshold values for the SVI subtheme scores may be of value from a public health perspective in tracking progress at the county level toward improved behavioral goals such as becoming more physically active. Interestingly, in linear regression modeling exclusively using SVI subtheme scores to predict physical inactivity prevalence, racial and ethnic minority status, which demonstrated the weakest relationship by correlation and ROC analyses, remained a significant predictor in the multivariate model. CDC data indeed indicate that physical inactivity prevalence in the United States is higher in racial/ethnic minority groups.³⁰ These diverging statistical findings, in addition to recent surveillance data, indicate that race/ethnicity is a meaningful factor in relation to

physical inactivity prevalence. As such, further work is needed to explore alternate methods of capturing racial/ethnic minorities in the context of measures of social vulnerability reflecting health behaviors such as physical inactivity.

In visualizing the distribution of SVI scores on a county level according to the American Nations model, clear heterogeneity was observed in the United States. Akin to the previously described “stroke belt,”³¹ Figure 2 provides evidence of a “social vulnerability belt” sweeping across a large portion of the southern US region—Alaska is also an area of concern. These findings are consistent with the distributions of unhealthy living behaviors and characteristics as well as chronic conditions we have previously demonstrated across the American Nations.¹⁴ Dichotomous grouping indicates clear differences in social vulnerability and physical inactivity prevalence (Table 5)—social vulnerability is not an equally shared experience across the United States. Moreover, this American Nations grouping added significant value to SVI subtheme scores in predicting physical inactivity prevalence. These findings collectively indicate: 1) social vulnerability and geographical cultural phenotypes may be interacting, with the latter impacting the presence of the former; 2) Both social vulnerability and geographic cultural phenotypes appear to be independently and synergistically impacting the prevalence of physical inactivity.

As mentioned previously, there has been little improvement in the physical inactivity crisis,⁵ which is a primary doorway to chronic disease risk and poor health outcomes. While well intentioned, the approach to addressing the physical inactivity crisis, from public health messaging to individual counseling, often takes a blanket, all-or-none approach: *Being physically active is good for everyone and hence all people should be more physically active, and exercising 150 minutes or more per week on most if not all days of the week at a moderate intensity is ideal.* While this overarching statement is certainly evidence based,^{27,28} little attention is paid to factors that influence a person’s decision to be physically active. Our findings indicate that social vulnerability and distinct US geographic regional cultures may both be influencing decisions to be physically active. These factors should be considered when designing physical activity health promotion campaigns and tailoring individual counseling—it’s time to apply a precision medicine approach to healthy living medicine.³²

Barriers to, and misperceptions of, physical activity should also be considered. For example, individuals who are unemployed and living below the poverty line likely face numerous barriers to becoming more physically active—the simple message of “you should exercise more” without strategizing on how to overcome such barriers is not an effective solution. Moreover, the message of “you should exercise at the ideal amount” may seem unattainable to a large percentage of the population, particularly those who are currently sedentary and experience social vulnerability. A qualitative study by Segar et al³³ evaluated “goals,

values, and daily priorities along with beliefs, feelings about and experiences being physically active” in White, Black, and Hispanic/Latina females who were grouped according to self-reported physical activity. Key findings included: 1) “low active participants tended to define physical activity in ways using a specific and narrow set of standards that traditional physical activity recommendations have used to educate people about physical activity”; 2) “low active participants also described feeling pressured by these criteria and not able to achieve them: You have to do this at this time, and you have to commit to these hours”; 3) “One low active participant said that *walking her dog was a barrier to being active*, which implied that she did not believe dog-walking counted as valid physical activity.”³³ The scientific evidence does not align with these perceptions. In truth, the greatest benefits in health trajectory are derived when an individual migrates from being sedentary to some level of physical activity, even well below the ideal goal.^{34,35} It is likely that the amount of physical activity needed to realize a health benefit is not understood by a majority of the population, a lack of understanding that is likely more prevalent in those who are physically inactive and socially vulnerable. Cultural influences on health behaviors must also be considered, and messaging should be tailored to resonate with cultural belief systems and values. Gaining a better understanding of the interactions of social vulnerability and culture and how they impact physical inactivity is essential to reverse the lack of progress in this area.

In conclusion, the findings of the current investigation indicate a crisis within a crisis with respect to the physical inactivity pandemic in the United States. Specifically, on a national level, physical inactivity prevalence is unacceptably high and has not appreciably improved over the past decade (Crisis 1). Within the United States, high levels of social vulnerability and physical inactivity are concentrated within specific geographic regions that need solutions to resolve this as well as other health disparities (Crisis 2).

References

- Arena R, McNeil A, Sagner M, Lavie CJ. Healthy living: the universal and timeless medicine for healthspan. *Prog Cardiovasc Dis* 2017;59(5):419–21.
- Sagner M, McNeil A, Puska P, et al. The P4 Health Spectrum - A Predictive, Preventive, Personalized and Participatory Continuum for Promoting Healthspan. *Prog Cardiovasc Dis* 2017;59(5):506–21.
- Sallis R. Exercise is medicine: a call to action for physicians to assess and prescribe exercise. *Phys Sportsmed* 2015;43(1):22–6.
- Sallis RE. Exercise is medicine and physicians need to prescribe it!. *Br J Sports Med* 2009;43(1):3–4.
- Arena R, Laddu D, Pronk NP, Woodard C. How has physical inactivity changed in the American Nations from 2010 to 2020: a decade of lost opportunity. *Curr Probl Cardiol* 2023;48(11):101965.
- Arena R, Hall G, Laddu DR, et al. A tale of one pandemic outliving another: Are even lower physical activity patterns following the COVID-19 pandemic the new norm?—A commentary. *Prog Cardiovasc Dis* 2023;79:53–5.
- Dunton GF, Do B, Wang SD. Early effects of the COVID-19 pandemic on physical activity and sedentary behavior in children living in the U.S. *BMC Public Health* 2020;20(1):1351.
- Zheng C, Huang WY, Sheridan S, Sit CH, Chen XK, Wong SH. COVID-19 pandemic brings a sedentary lifestyle in young adults: a cross-sectional and longitudinal study. *Int J Environ Res Public Health* 2020;17(17):6035.
- Tison GH, Barrios J, Avram R, et al. Worldwide physical activity trends since COVID-19 onset. *Lancet Glob Health* 2022;10(10):e1381–2.
- Andersen LB, Mota J, Di Pietro L. Update on the global pandemic of physical inactivity. *Lancet* 2016;388(10051):1255–6.
- Kohl HW 3rd, Craig CL, Lambert EV, et al. The pandemic of physical inactivity: global action for public health. *Lancet* 2012;380(9838):294–305.
- US Department of Health and Human Services, Office of Disease Prevention and Health Promotion, Healthy People 2030. Reduce the proportion of adults who do no physical activity in their free time—PA-01. Available at: <https://health.gov/healthypeople/objectives-and-data/browse-objectives/physical-activity/reduce-proportion-adults-who-do-no-physical-activity-their-free-time-pa-01>. 2023. Accessed August 28, 2023.
- Woodard C. *American Nations: A History of the Eleven Rival Regional Cultures of North America*. New York: Viking Press; 2011.
- Arena R, Laddu D, Pronk NP, Woodard C. The geographic distribution of unhealthy living characteristics according to the American Nations model: cultural factors warranting attention. *Prog Cardiovasc Dis* 2023;79:100–6.
- An R, Xiang X. Social vulnerability and leisure-time physical inactivity among US adults. *Am J Health Behav* 2015;39(6):751–60.
- Gay JL, Robb SW, Benson KM, White A. Can the social vulnerability index be used for more than emergency preparedness? An examination using youth physical fitness data. *J Phys Act Health* 2016;13(2):121–30.
- Jacobs MM, Burch AE. Disparities in perceived physical and mental wellness: relationships between social vulnerability, cardiovascular risk factor prevalence, and health behaviors among elderly US residents. *J Prim Care Community Health* 2023;14 [21501319231163639].
- Jain V, Al Rifai M, Khan SU, et al. Association between social vulnerability index and cardiovascular disease: a behavioral risk factor surveillance system study. *J Am Heart Assoc* 2022;11(15):e024414.
- Centers for Disease Control and Prevention (CDC). PLACES: local data for better health, county data 2023 release. Available at: <https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-20/swc5-unth>. 2023. Accessed September 8, 2023.
- Centers for Disease Control and Prevention (CDC). Behavioral Risk Factor Surveillance System. Available at: <https://www.cdc.gov/brfss/index.html>. 2023. Accessed September 26, 2023.
- Centers for Disease Control and Prevention (CDC)/Agency for Toxic Substances and Disease Registry (ATSDR). CDC/ATSDR Social Vulnerability Index. Available at: <https://www.atsdr.cdc.gov/placeand-health/svi/>. 2020. Accessed September 26, 2023.
- United States Census Bureau. American Community Survey (ACS). Available at: <https://www.census.gov/programs-surveys/acs>. 2022. Accessed September 27, 2023.
- Zelinsky W. *The Cultural Geography of the United States*. Hoboken, NJ: Prentice Hall; 1973.
- Arena R, Hall G, Laddu DR, Phillips SA, Lavie CJ. A tale of two pandemics revisited: Physical inactivity, sedentary behavior and poor COVID-19 outcomes reside in the same Syndemic City. *Prog Cardiovasc Dis* 2022;71:69–71.
- Booth FW, Roberts CK, Thyfault JP, Rueggsegger GN, Toedebusch RG. Role of inactivity in chronic diseases: evolutionary insight and pathophysiological mechanisms. *Physiol Rev* 2017;97(4):1351–402.
- Santos AC, Willumsen J, Meheus F, Ilbawi A, Bull FC. The cost of inaction on physical inactivity to public health-care systems: a population-attributable fraction analysis. *Lancet Glob Health* 2023;11(1):e32–9.

27. US Department of Health and Human Services. *Physical Activity Guidelines for Americans*. 2nd ed. Washington, DC: U.S. Department of Health and Human Services; 2018.; . Available at: https://health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf [Accessed November 13, 2023].
28. Bull FC, Al-Ansari SS, Biddle S, et al. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. *Br J Sports Med* 2020;54(24):1451–62.
29. Bevan G, Pandey A, Griggs S, et al. Neighborhood-level social vulnerability and prevalence of cardiovascular risk factors and coronary heart disease. *Curr Probl Cardiol* 2023;48(8):101182.
30. Centers for Disease Control and Prevention (CDC). Adult physical inactivity prevalence maps by race/ethnicity. Available at: <https://www.cdc.gov/physicalactivity/data/inactivity-prevalence-maps/index.html>. 2022. Accessed September 27, 2023.
31. Howard G, Howard VJ. Twenty years of progress toward understanding the stroke belt. *Stroke* 2020;51(3):742–50.
32. Arena R, Ozemek C, Laddu D, et al. Applying precision medicine to healthy living for the prevention and treatment of cardiovascular disease. *Curr Probl Cardiol* 2018;43(12):448–83.
33. Segar M, Taber JM, Patrick H, Thai CL, Oh A. Rethinking physical activity communication: using focus groups to understand women's goals, values, and beliefs to improve public health. *BMC Public Health* 2017;17(1):462.
34. Arena R, McNeil A, Street S, et al. Let us talk about moving: reframing the exercise and physical activity discussion. *Curr Probl Cardiol* 2018;43(4):154–79.
35. Lavie CJ, Lee DC, Sui X, et al. Effects of running on chronic diseases and cardiovascular and all-cause mortality. *Mayo Clin Proc* 2015;90(11):1541–52.